# DP-ShiftBench: How Distribution Shift Governs the Value of Pretraining for Private Learning

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## 1 Introduction

The world is moving towards pre-training with publicly available data and then to finetune on a private dataset using DP-SGD, which has been proven to be much more efficient and faster compared to training from scratch(Li et al. [2022], Ganesh et al. [2023], De et al. [2022]). However, the private and public datasets need not necessarily match, which is seen to be almost all the time, say for example pretraining on web images but finetuning on medical scans. But few studies actually show the effect of this distribution on the overall privacy–utlity tradeoff, does shift add to the privacy cost independently (Bassily et al. [2023]), amplify it by some factor, or is there some threshold post which the pretraining dataset has little to no effect and private data takes over the model outputs (Setlur et al. [2025])?

We propose **DP-ShiftBench**, a controlled benchmark where distribution shift is a *continuous and tunable parameter*. It features two tracks, images (SVHN→MNIST) and language (Markov chains), each evaluated across $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$ with $\delta = 10^{-5}$. Our contributions include: (1) a benchmark possessing fine grained shift control, (2) an evaluation of which divergence metrics (Renyi, KL, MAUVE Pillutla et al. [2023] to name a few) best analyse dataset variations under shift, and (3) a test of model predictive capability, trained with and without DP-SGD with respect to aforementioned divergences.

## 2 Proposed Approach

In both tracks, we pretrain on public data and finetune both with and without DP-SGD (Abadi et al. [2016]) on private data, sweeping $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$ ($\delta = 10^{-5}$). For each (pretrain, finetune) pair we compute specified divergences and plot a graph of model performance (for example perplexity) with respect to the divergence metrics.

### 2.1 Image Track: SVHN → MNIST

We pretrain on SVHN Netzer et al. [2011] (32×32 color digit images) and finetune with DP-SGD on MNIST. MNIST images are resised to $32 \times 32$ and replicated to 3 channels which account for the format mismatch. We use a SmallCNN (∼500K params) with GroupNorm. Table 1 summarizes the experiments: we control the shift via class-subset matching (pretrain on digits {0–4} vs. {5–9}) and pretraining objective (supervised vs. autoencoder). For each pair, we make sure that we compute FID and proxy $\mathcal{A}$-distance Ben-David et al. [2010] to quantify the magnitude of the shift.

| Experiment | Pretrain | Finetune (DP) | Shift Type |
|---|---|---|---|
| Baseline | None | MNIST | No pretrain |
| Full shift | SVHN (full) | MNIST | Domain gap |
| Matched | SVHN {0–4} | MNIST {0–4} | Domain only |
| Mismatched | SVHN {5–9} | MNIST {0–4} | Domain + label |
| Unsupervised | SVHN (autoenc.) | MNIST | Task gap |

Table 1: Image track experiments. Each run across $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$.

### 2.2 Language Track

**Markov chains.** We utilise a Markov chain of some order k ($k \in \{3, 4, 5\}$) to an existing dataset (say WikiText-2 or BooksCorpus), and create a transition matrix $[T]_P$, which depicts the distribution of our pretraining dataset $P$. We now slightly $[T]_P$ to create another transition matrix $[T]_Q$ and generate new sequences from $[T]_Q$ which will be used for finetuning, call this $Q$. Pertrubation is quantified using the Renyi divergence. We now pretrain a small language model on $P$ and finetune on $Q$ with both SGD and DP-SGD. The final step is to plot perplexity of the model with respect to the Renyi Divergence, for both SGD and DP–SGD, which is expected to be a monotonically increasing curve. The intuition behind doing this for SGD as well is to show that the increase in perplexity is not just because of the induced noise during the training process.

**Transformer LMs.** We use two complementary approaches: (1) generate $\geq$5,000 sequences (256–1024 tokens) from GPT-2 as pretraining data, then finetune on weight-perturbed variants; (2) use Pythia checkpoints at different training stages to naturally obtain shifted distributions. In both cases we measure the pretrain–finetune gap via MAUVE score Pillutla et al. [2023] and plot MAUVE vs. perplexity under SGD and DP-SGD.

**Math sequence models.** We construct controlled-vocabulary sequences where shift is an explicit parameter: (a) *noise perturbation*—randomly flip tokens by $\pm 1$ with probability $p$, so higher $p$ yields larger shift; (b) *arithmetic sequences*—pretrain on sequences with common difference $d_1$, finetune on $d_2 \neq d_1$; (c) *multiplicative sequences*—$s_{i+1} = m \cdot s_i + b \pmod{v}$, where increasing $m$ amplifies distributional change. These admit exact divergence computation and complement the natural-language experiments.

### 2.3 Analysis: Divergence Metrics and Interaction Hypotheses

For each (pretrain, finetune) pair across both tracks, we compute divergence metrics—FID and proxy $\mathcal{A}$-distance for images; Rényi divergence, KL, and total variation for language—and correlate them with DP finetuning performance. We fit three interaction models to the full 2D grid ($\varepsilon \times$ shift, 3 seeds each): $H_{\text{add}}$: Err $= C_1/\varepsilon + C_2 d + C_3$ Bassily et al. [2023]; $H_{\text{mult}}$: Err $= C_1 d/\varepsilon + C_2$; $H_{\text{thresh}}$: Err $= C_1/\varepsilon + C_2 d$ if $d < d^*$, else $C_1/\varepsilon + C_3$ Setlur et al. [2025]. We compare fits via AIC/BIC to determine which model best describes the shift–privacy interaction.

## 3 Relevant Literature

**Pretraining for DP.** Li et al. [2022] showed that pretraining dramatically improves DP learning for LLMs. Ganesh et al. [2023] provides theoretical backing via a two-phase analysis to the fact that public pretraining is necessary. De et al. [2022] shows that scale combined with pretraining unlocks high DP accuracy on image classification. Tramèr and Boneh [2021] proved the bottleneck under DP is a feature quality and not optimization.

**Distribution shift under DP.** Bassily et al. [2023] derived additive error bounds for differentially private domain adaptation. Setlur et al. [2025] validates the threshold intuition, by defining a Goldilocks zone for proving lower bounds which essentially says that there is a critical threshold for the shift above which public data stops having an effect . Ben-David et al. [2010] provide the foundational domain adaptation theory ($\mathcal{H}$-divergence, proxy $\mathcal{A}$-distance) that we will be building upon.

**Measuring distribution gap.** We quantify the pretraining–finetuning gap using the FID metric ( Parmar et al. [2022]), proxy $\mathcal{A}$-distance (Ben-David et al. [2010]), and the divergence frontier approach of Pillutla et al. [2021, 2023] (MAUVE).

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